

ITAM

Financial Models I

Course Description:

This is an advanced undergraduate course in finance. The course focuses on real world market risk models for equity stocks and equity options. Working on real life models, students develop their intuition in finance matters and learn derivatives pricing, risk management and market model implementation. This course prepares students to work in Risk Management for any financial institution.

Objectives:

In this course you will develop the following skills:

1. Apply statistics and mathematics to real financial problems
2. Read and write financial problems (from the formula on paper to Excel)
3. Problem solver and problem implementation: Learn advanced finance concepts and how to solve them in Excel
4. Become an expert of the equity market and its risk management tools

Prerequisites:

Basic knowledge of Excel and option contracts is desirable, but not essential. Basic knowledge in statistics and econometrics.

Assessment:

- **Attendance (10%):** based on class attendance
- **Final Project (5%):** to be done in groups of two
- **Assignments (5%):** to be done in groups of two.
- **Midterm (30%):** The examination is open book in Excel.
- **Final Exam (50%):** comprehensive exam of all material covered

Academic Integrity

*ITAM values academic integrity. Therefore, all students must understand the meaning and consequences of cheating, plagiarism and other academic offences under the Code of Student Conduct. **ZERO TOLERANCE TO CHEATING, COPYING OR PLAGIRISM.***

Tentative Lectures Schedule:

1. Risk Management and Financial Returns (Ch. 1)
2. Historical Simulation, Value at Risk (VaR) and Expected Shortfall (Ch. 2)
Backtesting and Stress Testing
3. Volatility Modeling using daily data (Ch. 3 with special emphasis on Section 4 & Ch. 4)
4. Volatility Modeling using intraday data (Ch. 5)
5. Nonnormal distributions(Ch. 6)
6. Correlation Modeling (Ch. 7)
7. Term structure of risk (Ch. 8)
8. Copulas (Ch. 9)
9. Option Pricing (Ch. 10)
10. Option Risk Management (Ch. 11)
11. Credit Risk Management (Ch. 12)

Required Readings

Elements of Financial Risk Management by Peter Christoffersen. Second Edition 2012.

Additional Literature

Options, Futures, and Other Derivatives by John Hull.
Value at Risk by Philippe Jorion.